www.covenantuniversity.edu.ng

PHY211



Covenant University

Raising a new Generation of Leaders

ELEMENTARY MODERN PHYSICS

BY:

BABARIMISA O. IDOWU

OFFICE NO: ROOM 187, LECTURE THEATRE

Experimental Basis of Quantum Theory or Mechanics

Why Quantum Mechanics (QM)?

It evolved due to the limitations of classical or Newtonian mechanics to explain small sized particles.

QM is an application of quantum mechanics to physical systems.

QM is the study of the mechanics or laws of motions of microscopic particles: atoms, subatomic particles (of diameter ~ 10^{-12} m).

QM was introduced towards the end of 19th century because classical mechanics fell short in describing systems composed of small particles.

Newtonian / Classical Mechanics

Newtonian Mechanics is a branch of science based on Newton's laws of motion and Maxwell's electromagnetic wave theory to explain the phenomena associated with wave's motion and energy.

Classical Mechanics was introduced in the 17th century and is successful in explaining our every day experience.

The Failures of Classical Mechanics

According to classical explanation:

1. **The locations and momenta of small moving objects can be specified simultaneously with a higher precision which is incorrect.**
2. **Nature was composed of two separate types of entities which are called particles and waves. Matter was treated as a particle and light was treated as a wave, however, they are not separate entities.**
3. **Energies involved in the absorption and emission of electromagnetic radiation vary *continuously* (physically observable variables can take on a continuum values).**

Classical Mechanics explained successfully the mechanics of macroscopic particles only

It is failed to describe the motions of individual atom and subatomic particle.

The basic principles of classical mechanics:

1. It predicts a precise trajectory for particles, with precisely specified locations and momenta at each instant.

2. It permits the translational, rotational, and vibrational modes of motion to be excited to any energy simply by controlling the forces that are applied.

The above conclusions agree with daily experience

The daily experience, however, does not extend to individual atoms, and careful experiments have shown that classical mechanics fails when applied to the transfers of very small energies and to objects of very small mass.

Therefore, physicists of the last 19th century were faced with a great challenge and tried to examine classical mechanics by performing different careful experiments and they tried to originate and evolve the concept of quantum mechanics, at least in principle to predict behavior of the universe.

Why are quantum mechanics basic in Physics?

Quantum mechanics predicts what Classical mechanics could not, at the level of subatomic particles:

the binding of electrons inside a hydrogen atom,

quantum tunneling,

and other phenomena.

Applications of quantum mechanics (QM):

It is an application of QM to solve problems relating to physical processes:

To understand structure of individual atoms

To understand interatomic forces

To know how subatomic particles move in reaction to the forces that they experience

To interpret atomic spectra

To evaluate thermodynamic properties,

Experimental Foundation of Quantum Mechanics

EXPERIMENTS LEADING TO THE CONCEPT OF QUANTUM MECHANICS:

1. Black body radiation
2. Photoelectric Effect
3. The Compton Effect
4. The Line Spectra of Atoms
5. The Rutherford model of Hydrogen atom
6. The Bohr Model of Hydrogen atom
7. Wave-particle duality of Light
8. The Wave Properties of Particles

The Schrödinger Equation

Schrödinger wave equation is a basic mathematical expression of wave mechanics which describes the behavior of one or more particles.

It considers the 2 basic laws that govern the behavior of electrons in atoms:

- **De Broglie's relation**
- **Heisenberg uncertainty principle**

The 2 laws show the wave-particle duality of particles and the probable distribution of electrons instead of the well defined orbits around the nucleus of an atom as earlier postulated by Neil's Bohr.

Black body radiation

When an object is heated, there is a continual shifts of color from dull red to red then to white
as temperature increases more and more.

This behavior can be seen from the behavior of hot objects called black body:

A black body is an idealized body that is capable of absorbing and emitting all frequencies of radiation uniformly.

Black body radiation is the radiation emitted by a **black body**

The **intensity** of the emitted radiation depends on:

- The temperature of the body and
- the wavelength or frequency of the radiation.

Blackbody radiation does not depend on the type of object emitting it.
The entire spectrum of **blackbody radiation** depends on only one parameter, the **temperature, T**

Photoelectric Effect

Photoelectric effect is the ejection of electrons from the surface of a metal by irradiating ultraviolet (UV) radiation on the metal.

The **Photoelectric effect** demonstrates the **particle nature** of light as discovered by **Albert Einstein**

The Compton Effect

Physicist A. H. Compton, studied the scattering of electromagnetic radiation (EMR) from various materials.

He observed that after the scattering, the frequency (energy) of the EMR had changed, (had *reduced*).

When a high energy photon collides with an electron that is stationary, both particles may be scattered at an angle to the direction of the path of the incident radiation and this was observed experimentally.

If photons are particles, they should have momentum so that a collision between the photons and the electrons is possible.

This was observed experimentally by allowing a beam of light to fall on a stream of electrons,
a transfer of momentum would be observed as a scattering of electrons by the beam of light (photons).

Compton effect (or Compton scattering) can be understood by assuming that the scattering process is a photon - electron collision in which momentum and energy are conserved.

By classical assumption,
The wavelength after scattering should be the same before scattering

However, at a given scattering angle,
it was shown that only one value of wavelength shift is observed,
thus momentum transfer took place in a discrete manner which shows radiation is a stream of particles (photons).

The Line Spectra of Atoms

- **Spectrum is a continuous distribution of colored light produced when a beam of white light is separated into its components**
- **Spectroscopy is a science that deals with spectra to determine the physical properties and chemical composition of substances.**
- **Spectroscope is a device for producing and observing a spectrum.**
- **Spectrograph is a device for observing and recording a spectrum *photographically*.**

- **Spectrophotometer is an instrument for measuring spectrum wavelength intensities**
- **Line Spectrum is a spectrum of distinct lines characteristics of emission or absorption by atoms**

  It is a spectrum made up of series of distinct lines which are characteristics of emission or absorption of light by atoms.

  The theory of quantization led to the interpretation of atomic spectral lines from the study of light.

  For several years, scientists tried to find some pattern in the frequencies of lines in atomic hydrogen-atoms

  (*Hydrogen is the lightest and simplest atom which has simplest spectrum so easy to study*).

Wave-particle duality of Light

- Light can be characterized by wavelength and frequency, which indicates its wave nature and

- light can be characterized or expressed by energy and momentum, which indicates its particle nature.

Hence, light has both wave and particle nature, these behaviors of light is called wave-particle duality

The Wave Properties of Particles

In 1924, Louis de Broglie proposed a new speculative hypothesis that electrons and other particles of matter can behave like waves.

Today, this idea is known as de Broglie's hypothesis of matter waves.

In 1926, De Broglie's hypothesis, together with Bohr's early quantum theory, led to the development of a new theory of wave quantum mechanics to describe the physics of atoms and subatomic particles.

- **Quantum mechanics has paved the way for new engineering inventions and technologies, such as:**
- **the laser and**
- **magnetic resonance imaging (MRI).**

These new technologies drive discoveries in other branches of science such as biology and chemistry.